

LINEAR PREDICTORS

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Learning Objectives

1. Linear Predictors
2. **Linear Regression**
3. Loss for Actual vs Predicted
4. Cost Function for Linear Regression
5. Parameters Impact on Cost Function
6. **Logistic Regression**
7. Cost Function for Logistic Regression
8. Non-Linearity for Regression
9. Polynomial Regression

Introduction

The first family of hypothesis classes we will explore is linear predictors, including **Linear Regression** and **Logistic Regression** predictors.

A linear predictor function is a **linear function** (linear combination) of a set of **coefficients** and **explanatory variables** (independent variables), whose value is used to predict the outcome of a dependent variable

X and Y grow in Linear fashion

Introduction

Many learning algorithms that are being widely used in practice rely on Linear Predictors.

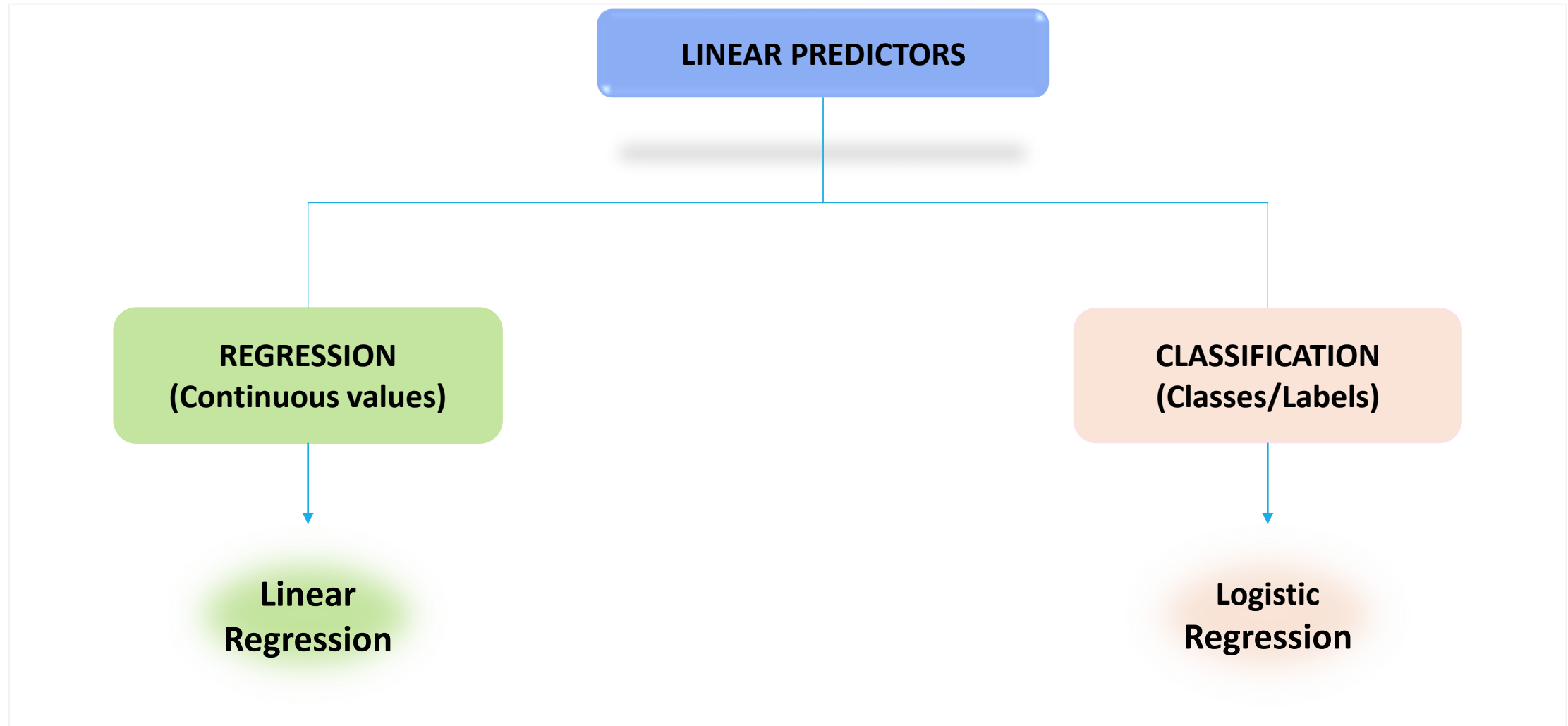
Linear Predictors have the ability to **learn from datasets efficiently** in many cases.

They are intuitive, easy to interpret, and **fit the data reasonably well** in many natural learning problems.

Linear Predictors have a set of functions that are **parameterized by weights and bias**.

These functions take input **X**(vector) and return output **Y**(scalar) in the form of a linear function $\langle \text{weights}, x \rangle + \text{bias}$.

Linear Predictors



How Linear Predictors Learn?

The output Y is a function of input X .

$$Y = \text{function of } x$$

- $Y = g(x|\Theta)$ Here, g is fixed and is a **Linear predictor** & ' Θ ' is a parameter

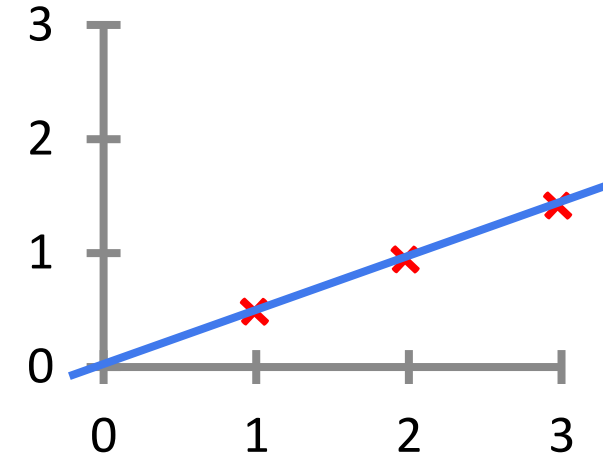
Given a problem statement, we will frame the Hypothesis to predict the results.

- Hypothesis $h(x) = (x|\Theta)$

Linear Predictors learn the relationships between the independent variables(input) $x_1, x_2, x_3, \dots$ and the dependent variable(output) Y .

Linear Relationship

X (input/feature)	Y (output/target)
1	0.5
2	1
3	1.5



The equation for the Straight Line is given by

$y = mx + c$ $\longrightarrow$ 'm' is the slope and c is the y-intercept(constant)

$h_{\theta} = \theta_1 x + \theta_0$ $\longrightarrow$ ' θ_1 ' is the weight(slope) and θ_0 is the bias term(constant)

$y = w.x + b$

Linear
Predictors

Linear Regression

Linear Regression is a standard statistical tool for modeling the relationship between some input independent variables X and real-valued outcome dependent variable Y .

Here output Y is the **Linear function** of input X . ($X = x_1, x_2, x_3, \dots$)

The Linear Model is a **straight line** that fits the dataset in the best possible way.

Linear Regression learns patterns from the dataset using **parameters called weight and bias**.

Linear Regression

$X_1 \rightarrow \text{Feature1}$

	X_1	X_2	X_3	X_4	...	...	Y
X^1							
X^2							
X^i							

Training Set

$X^1 \rightarrow \text{Datapoint1}$
(instance)

Test Set

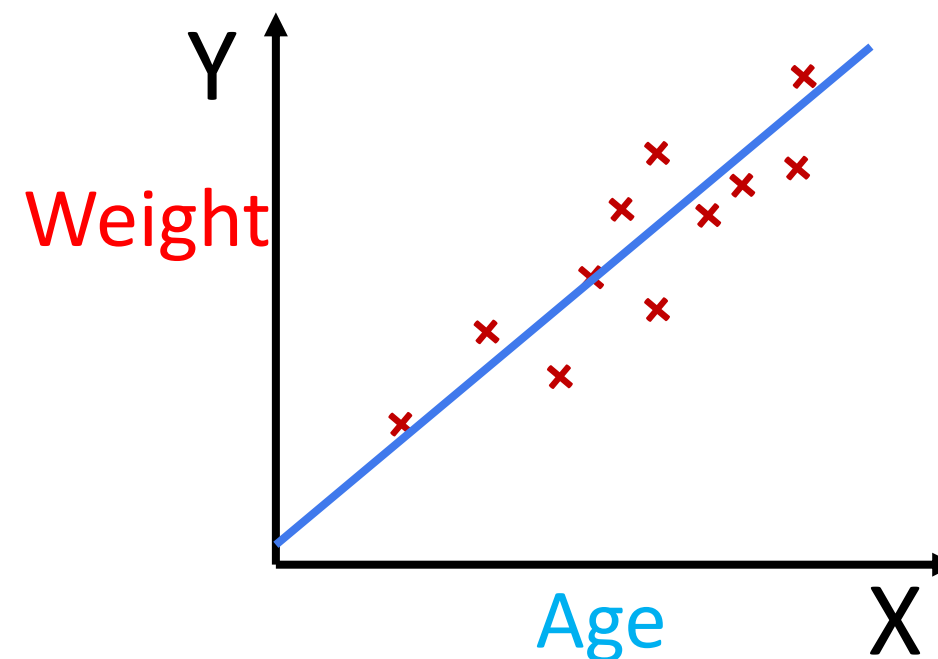
	X	...	...	...	Y

MODEL
(g-Linear)

Unseen
Age X

HYPOTHESIS (h)

Predict
Weight $\hat{y}$

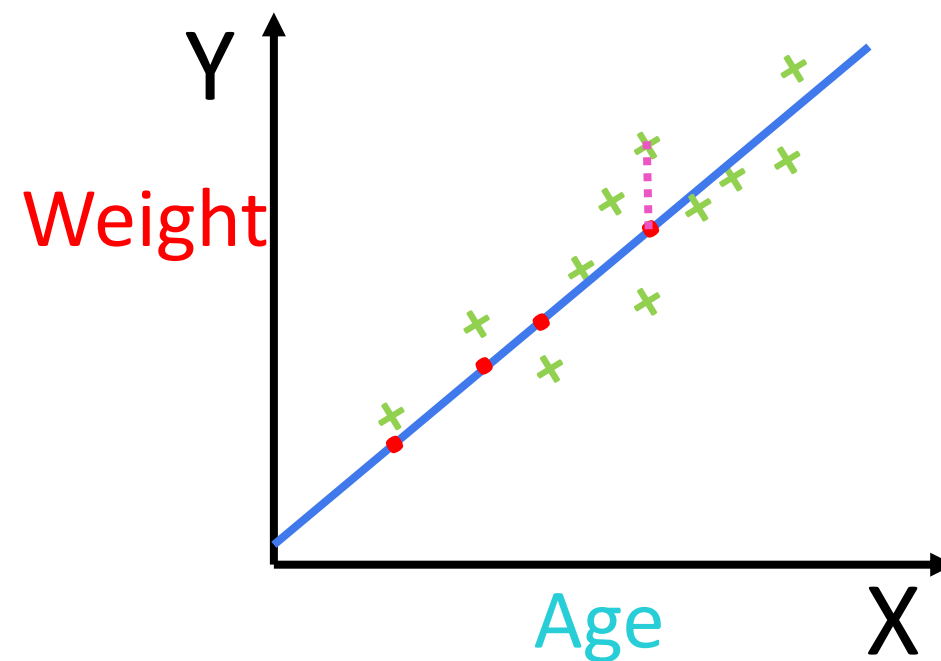


Simple Linear Regression

- In Linear Regression we always find the best fit line so that it approximates the relationship between independent and dependent variables.
- To find the final best-fit line start with first data point and aim for minimal distance between actual and predicted points.
- Linear Regression with one variable is called Univariate Linear Regression.

Hypothesis:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



Also expressed as $\hat{y} = WX + b$

Linear Regression

Consider another example of housing prices prediction. The independent variables are :

1. ($\mathbf{x}_1$) area of the house in square feet,
2. ($\mathbf{x}_2$) no. of bedrooms,
3. ($\mathbf{x}_3$) location

X_1	X_2	X_3	X_4	...	...	Y

In general, when designing a learning problem for different problem statements the domain experts(SME) will help to decide which features are significant

$$\mathbf{h}_{\theta}(\mathbf{x}) = \theta_0 + \theta_1 \mathbf{x}_1 + \theta_2 \mathbf{x}_2 + \theta_3 \mathbf{x}_3 \dots$$

Here, the θ_i 's are the parameters (also called coefficients or weights) and $i=1,2,3\dots$ parameterizing the space of linear functions mapping from X to Y .

To simplify, $\mathbf{h}_{\theta}(\mathbf{x})$ is denoted as $\mathbf{h}(\mathbf{x})$

Simple Linear Regression – Example

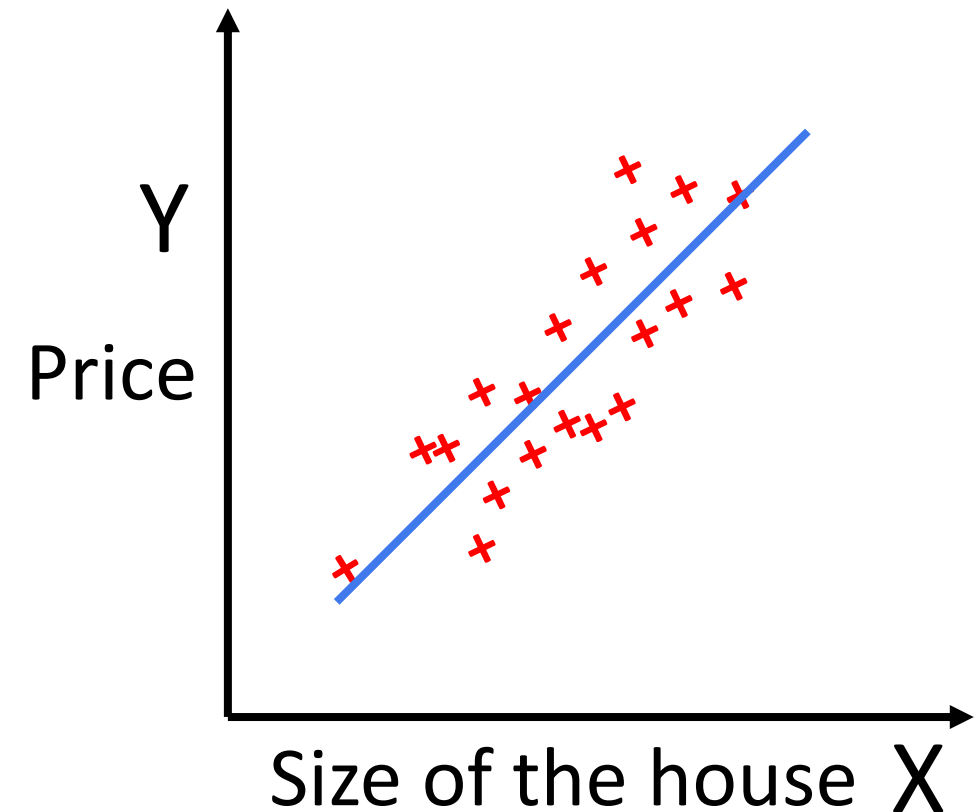
Another example of the Simple (**Univariate**) Linear Regression is predicting the house prices based on the size of the house.

Linear Regression

Hypothesis:

$$h(x) = \theta_0 + \theta_1 x$$

Size in feet ² (x)	Price (\$) in 1000's (y)
2023	445
1115	223
1234	307
625	166
...	...



Linear Datasets

Size in feet ² (x)	Price (\$) in 1000's (y)
500	125
1000	215
750	155
1500	330
800	170
1250	235
1400	295
650	135
900	250
1150	205

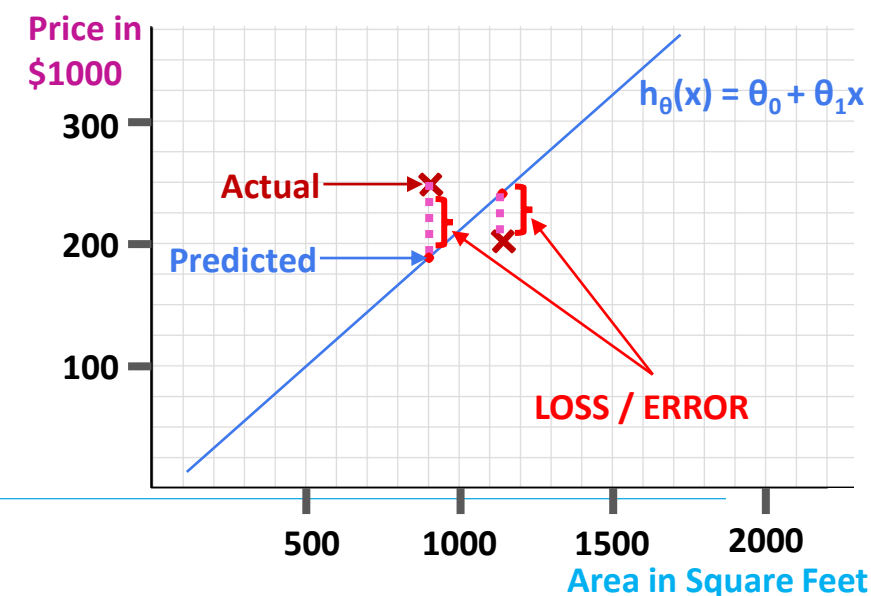
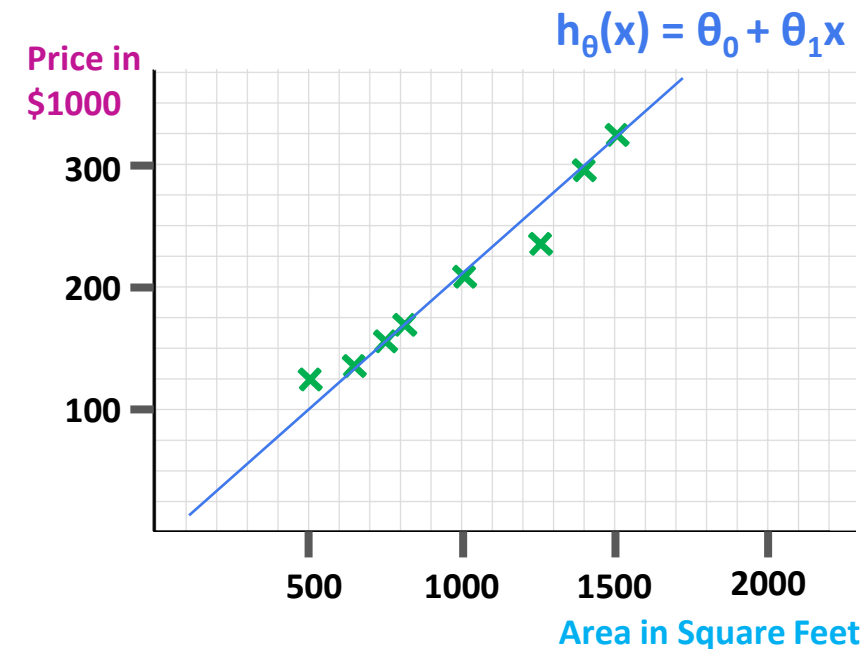
m = sample size **10**

80% **Training Set** sample size = **8**

Size in feet ² (x)	Price (\$) in 1000's (y)
500	125
1000	215
750	155
1500	330
800	170
1250	235
1400	295
650	135

20% **Non-Training** sample size= **2**

Size in feet ² (x)	Price (\$) in 1000's (y)
900	250
1150	205



Linear Regression

- To simplify our notation, we also introduce the convention of letting $\theta_0 = 0$ (this is the intercept term), so that

$$\mathbf{h}(\mathbf{x}) = \sum_{i=0}^d \theta_i \mathbf{x}_i = \boldsymbol{\theta}^T \mathbf{x}$$

- Where on the right-hand side above we consider $\boldsymbol{\theta}$ and $\mathbf{x}$ both as **vectors**, and here d is the number of input variables.
- Now, given a training set, how do we pick, or learn, the parameters θ_i 's?
- One reasonable method is to make the ' $\mathbf{h}(\mathbf{x})$ '-predicted value($\hat{\mathbf{y}}$), close to ' $\mathbf{y}$ '-actual value, at least for the training examples.

Vector notation

$$h(x) = \theta^T x + \theta_0$$

$$h(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$$

$$X = [x_1, x_2, x_3 \dots x_n]_{1 \times n}$$

$$\theta = [\theta_1, \theta_2, \theta_3, \dots, \theta_n]_{1 \times n}$$

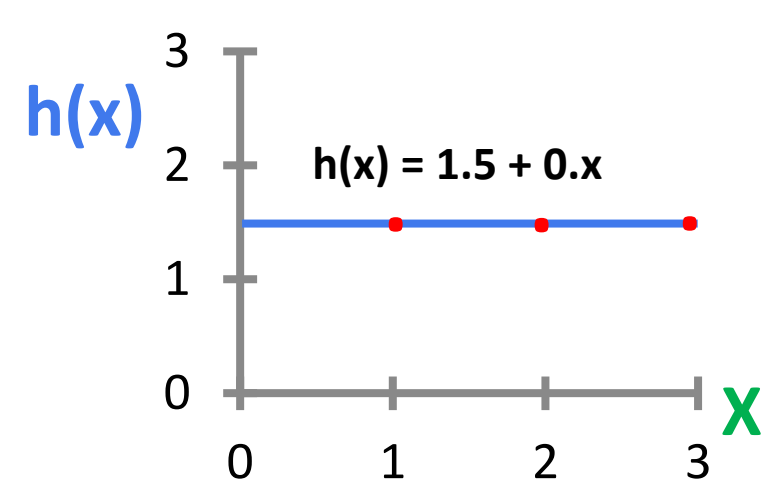
$$\theta \rightarrow \theta^T$$

$$\theta^T = \begin{bmatrix} \theta_1, \\ \theta_2, \\ \theta_3, \\ \vdots \\ \theta_n \end{bmatrix}_{n \times 1}$$

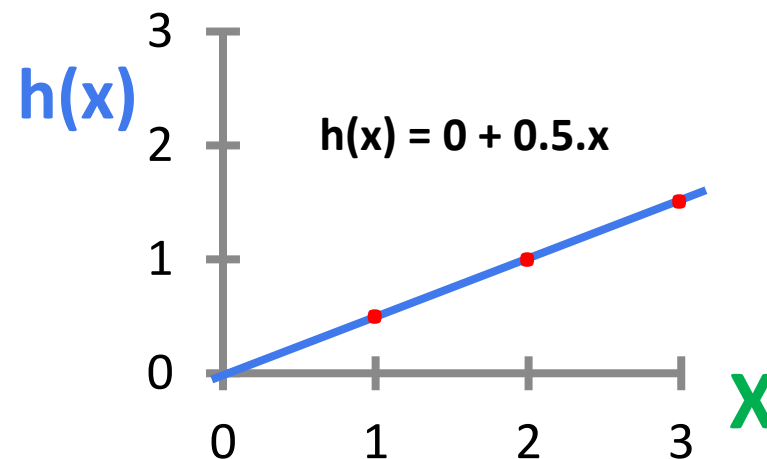
$$X = [x_1, x_2, x_3 \dots x_n]_{1 \times n}$$

Parameters Impact on Linear Predictor

$$h(x) = \theta_0 + \theta_1 x$$

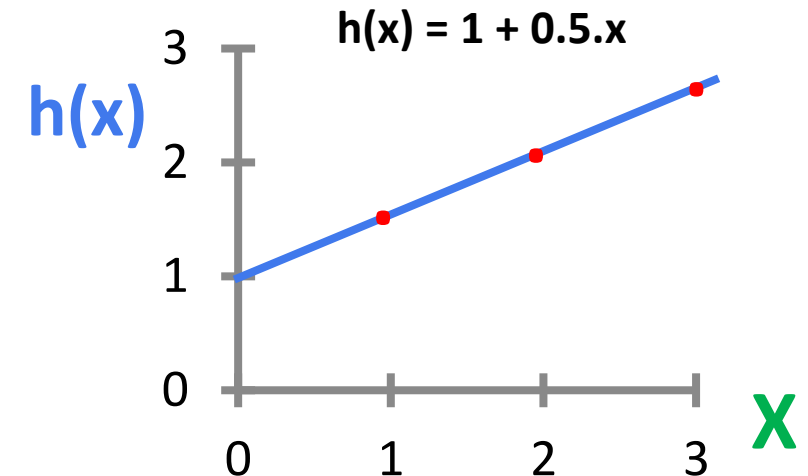


x	$\theta_0 = 1.5$
1	$\theta_1 = 0$
2	
3	



$$\theta_0 = 0$$

$$\theta_1 = 0.5$$



$$\theta_0 = 1$$

$$\theta_1 = 0.5$$

How to choose parameters θ_i 's (or weights or co-efficients) ?

Loss – Actual vs Predicted

- In regression, if the baby's weight is 3 kg, both the predictions **3.00001** kg and 4 kg are “wrong,” but we would clearly prefer the former over the latter.
- We therefore need to define how much we shall be “**penalized**” for the discrepancy between **$h(\mathbf{x})$** (the predicted value) and **y** (expected or actual value).

- One common way is to use the **Squared-Loss** function,

$$l(h(\mathbf{x}, y)) = (h(\mathbf{x}) - y)^2 \quad (\text{predicted} - \text{actual})^2 \quad l(h(\mathbf{x}, y)) = (\hat{y} - y)^2$$

- For this loss function the empirical risk function is called **Mean Squared Error**

$$L_s(h) = 1/m \sum_{i=1}^m (h(\mathbf{x}^i) - y^i)^2$$

- The loss function is also referred to as **Cost Function** or **Error**.

Cost Function for Linear Regression

The final goal is to find the optimal values of the **intercept** θ_0 and **slope** θ_1 of the line, that best fits the data and minimizes the distance between the data points and the line.

To make $\mathbf{h}(\mathbf{x})$ close to $\mathbf{y}$, for the training examples we will define a function that measures:

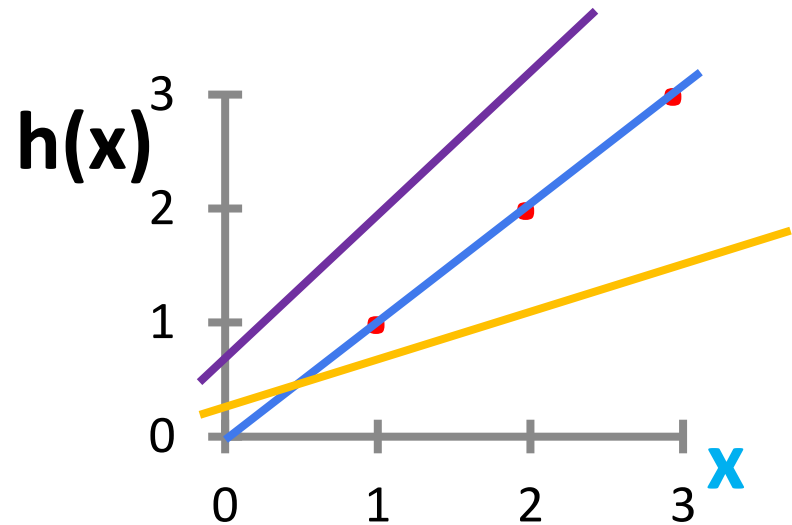
For each value of the θ 's, how close the $\mathbf{h}(\mathbf{x}^{(i)})$'s are to the corresponding $\mathbf{y}^{(i)}$'s.

We define the cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\mathbf{h}(\mathbf{x}^i) - \mathbf{y}^i)^2$$

Parameters Impact on Hypothesis

Hypothesis $h(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$



x
1
2
3

$\theta = 1.0 \quad h(x) = 1.x$

$\theta = 0.5 \quad h(x) = (0.5).x$

$\theta = 1.5 \quad h(x) = (1.5).x$

When we compute $h(x) = \theta x$ for each value of θ and X we get the following results

x	h(x)	h(x)	h(x)
1	1	0.5	1.5
2	2	1	3
3	3	1.5	4.5

Next, use these predicted values to estimate the cost

Parameters Impact on Cost Function

Cost Function $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2$

Sample size $m=3$ (no. of input values 'x')

$h(x)$ θ

$$J(\theta) = \frac{1}{2m} ((h(x^1) - y^1)^2 + (h(x^2) - y^2)^2 + (h(x^3) - y^3)^2)$$

predicted values

x	h(x)	h(x)	h(x)
1	1	0.5	1.5
2	2	1	3
3	3	1.5	4.5

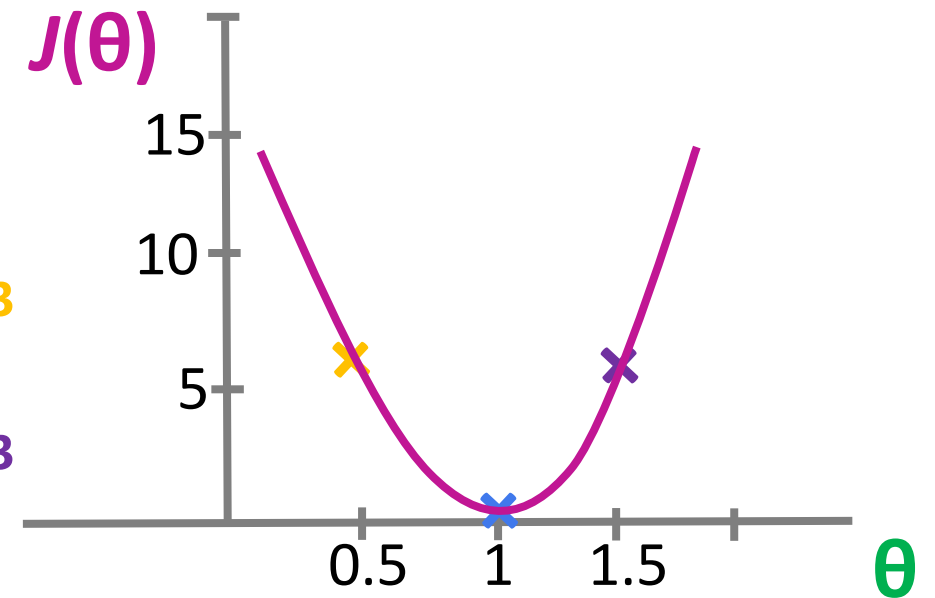
$h(x^1)$

$h(x^2)$

$h(x^3)$

Actual(expected) values

y
1
2
3



Later Add Bias Term

$\theta = 1.0$ $h(x)$ $J(1) = \frac{1}{2(3)} ((1-1)^2 + (2-2)^2 + (3-3)^2) = 0$

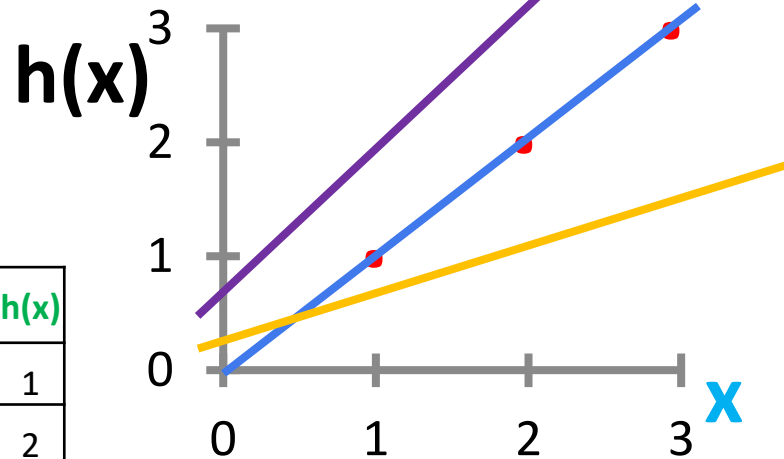
$\theta = 0.5$ $J(0.5) = \frac{1}{2(3)} ((0.5 - 1)^2 + (1 - 2)^2 + (1.5 - 3)^2) = 0.583$

$\theta = 1.5$ $J(0.5) = \frac{1}{2(3)} ((1.5 - 1)^2 + (3 - 2)^2 + (4.5 - 3)^2) = 0.583$

Summary of Parameters Impact on H & Cost Function

Hypothesis $h(x) = \sum_{i=0}^d \theta_i x_i = \theta^T x$

x	h(x)
1	1
2	2
3	3

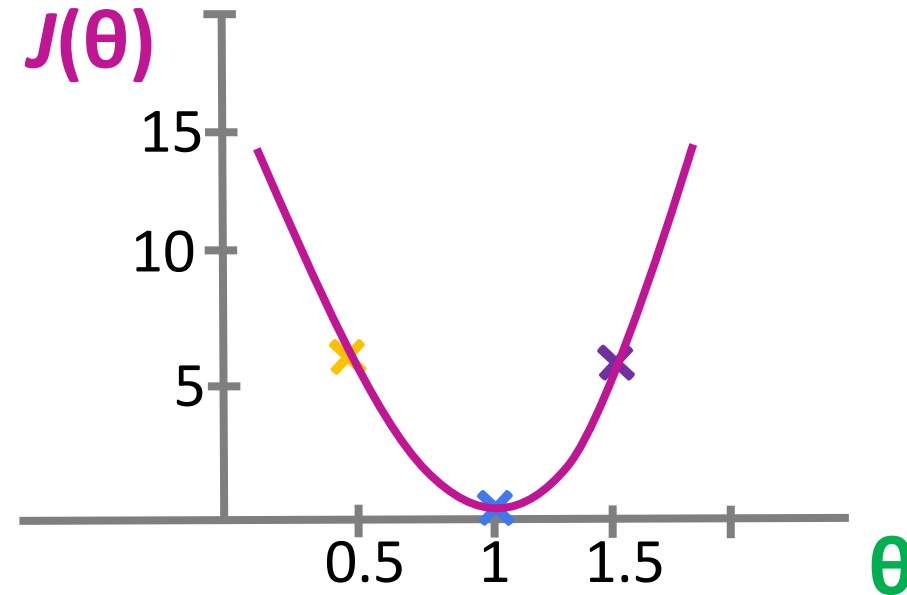


$\theta = 1.0 \quad h(x) = 1.x$

$\theta = 0.5 \quad h(x) = (0.5).x$

$\theta = 1.5 \quad h(x) = (1.5).x$

Cost Function $J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2$



y
1
2
3

$\theta = 1.0 \quad h(x) \quad J(1) = \frac{1}{2(3)} ((1-1)^2 + (2-2)^2 + (3-3)^2) = 0$

$\theta = 0.5 \quad J(0.5) = 0.583$

$\theta = 1.5 \quad J(1.5) = 0.583$

Later Add Bias Term

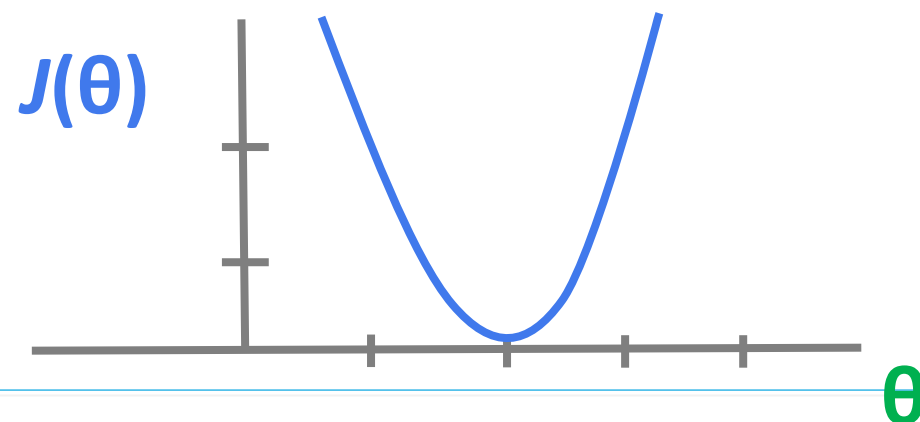
Optimizing Cost Function

The Linear Regression model must have a **minimum cost function** for more accurate prediction.

This minimum cost is **achieved by adjusting the parameters** – bias and weights.

The minimum value of the cost function determines how **closely the predicted and actual values are**.

Hence **optimizing the Cost Function is critical** and significant to develop an **efficient model** with good predictions.



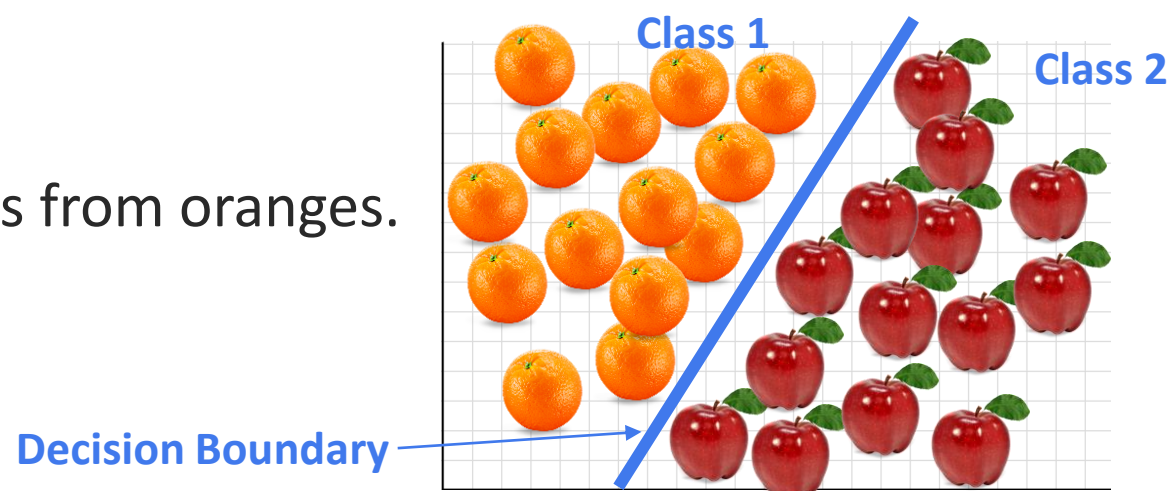
Decision Boundary

The Decision Boundary separates the data points into specific classes, where the algorithm switches from one class to another.

On one side of a decision boundary, all datapoints are more likely to be called as class 1 and on the other side of the boundary, it's more likely to be called as class 2 if the problem is Binary Classification.

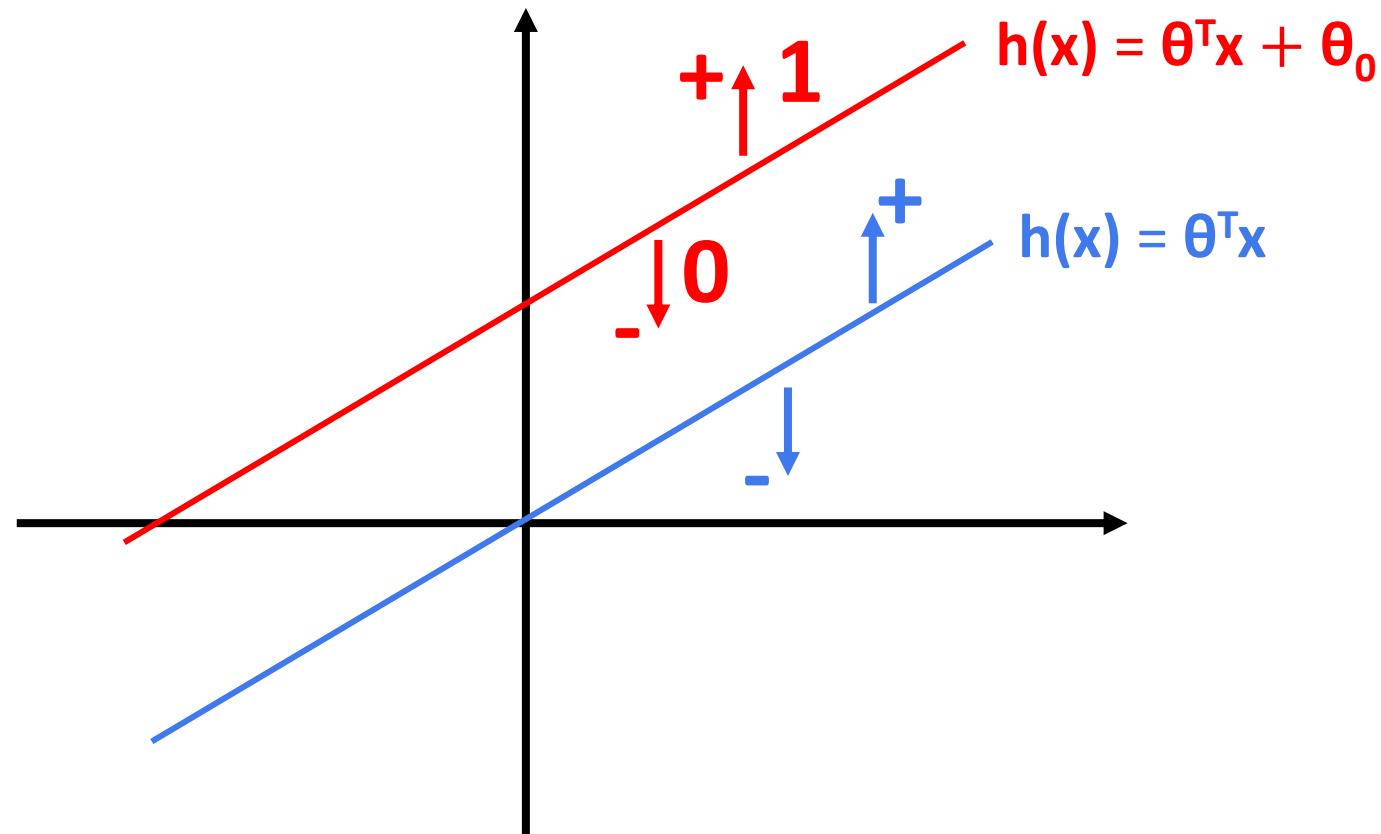
The decision boundary is different for datasets that can be either linear datasets or non-linear datasets

Example a line that can separate apples from oranges.



Half Spaces

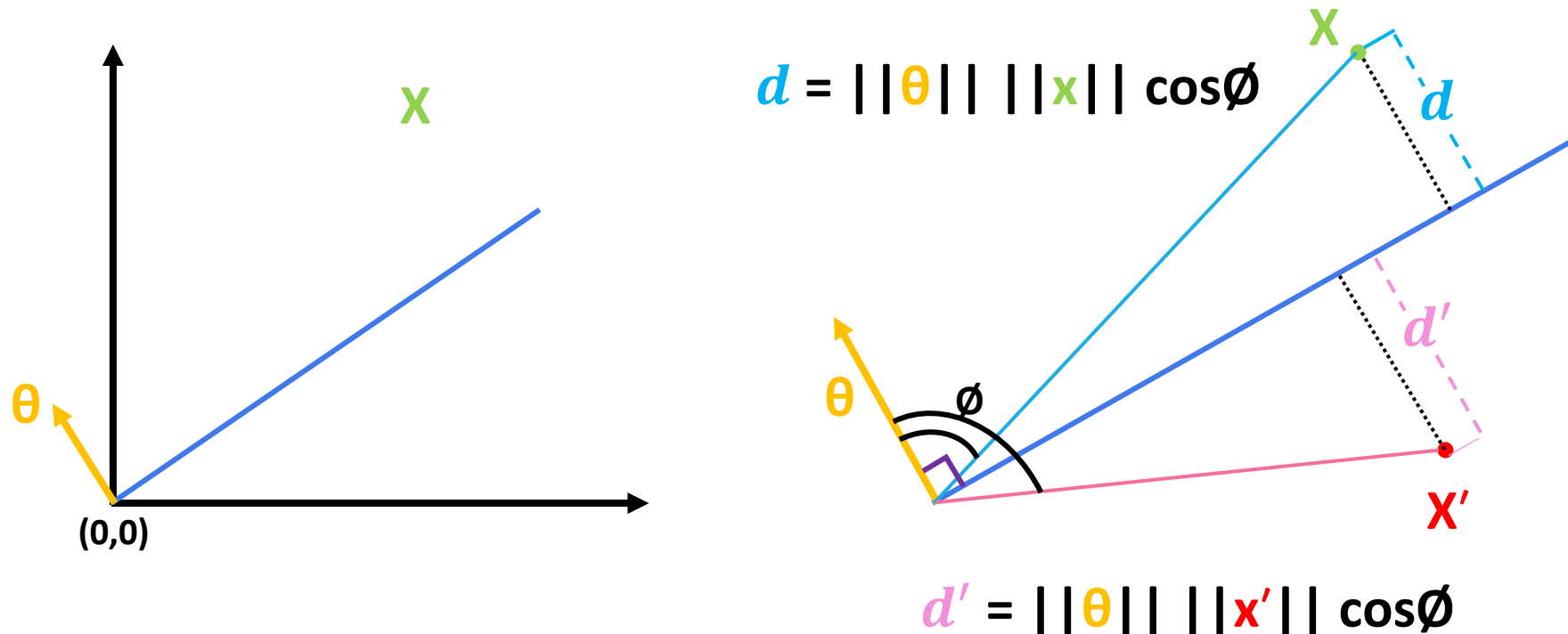
The hypothesis class we consider is the class of half-spaces. These are designed for **binary classification** problems.



Half Spaces

Understanding the distance of a point from the line (decision boundary)

distance of the point d = dot product of (θ, x)



ϕ	$\cos \phi$
0	$\cos 0^\circ = 1$
30	$\cos 30^\circ = \sqrt{3}/2$
45	$\cos 45^\circ = 1/\sqrt{2}$
60	$\cos 60^\circ = 1/2$
90	$\cos 90^\circ = 0$
120	$\cos 120^\circ = -1/2$
150	$\cos 150^\circ = -\sqrt{3}/2$
180	$\cos 180^\circ = -1$
270	$\cos 270^\circ = 0$
360	$\cos 360^\circ = 1$

Logistic Regression

Logistic regression is used for **Classification Tasks**.

Email: Spam / Not Spam? (Yes / No)

Fruit: Apple/ Orange? (Yes / No)

Online Transactions: Fraudulent (Yes / No)?

Tumor: Malignant / Benign? (Yes/No)

Animal: Cat/Dog (Yes/No)

In these problems, the variable that we want to predict can only be one of two possible values: No or yes.

This type of classification problem where there are only two possible outputs is called **Binary Classification**

$$\mathbf{y} \in \{0, 1\}$$

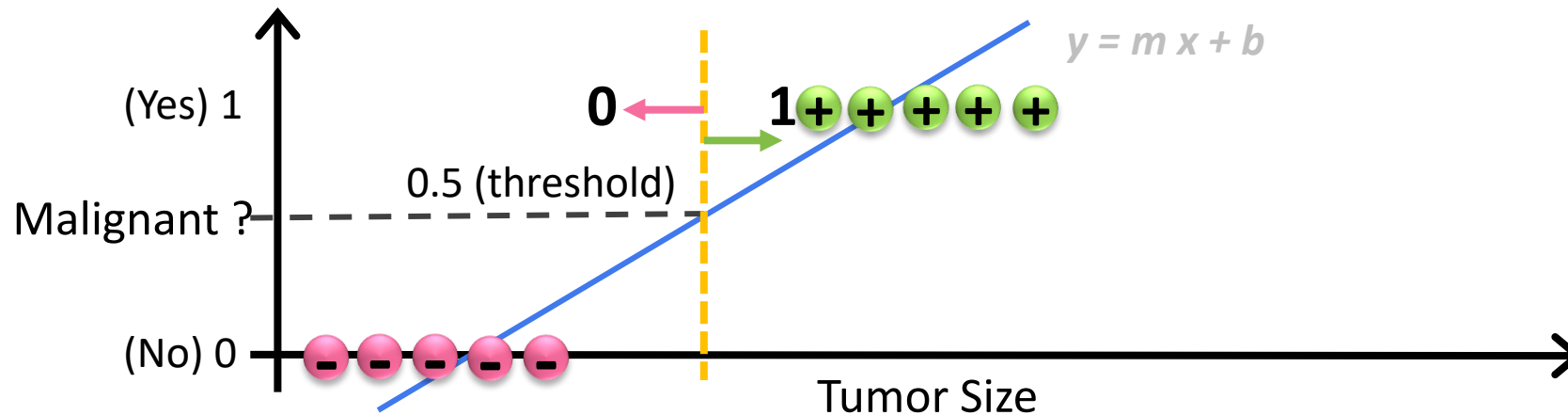
0: “Negative Class” (e.g., benign tumor)

1: “Positive Class” (e.g., malignant tumor)

Linear Regression for Classification

Is it appropriate to use Linear Regression for Classification?

Consider an example of the classification of tumors.



Threshold classifier output $h_{\theta}(x)$ at 0.5:

If $h_{\theta}(x) \geq 0.5$, predict “y = 1”

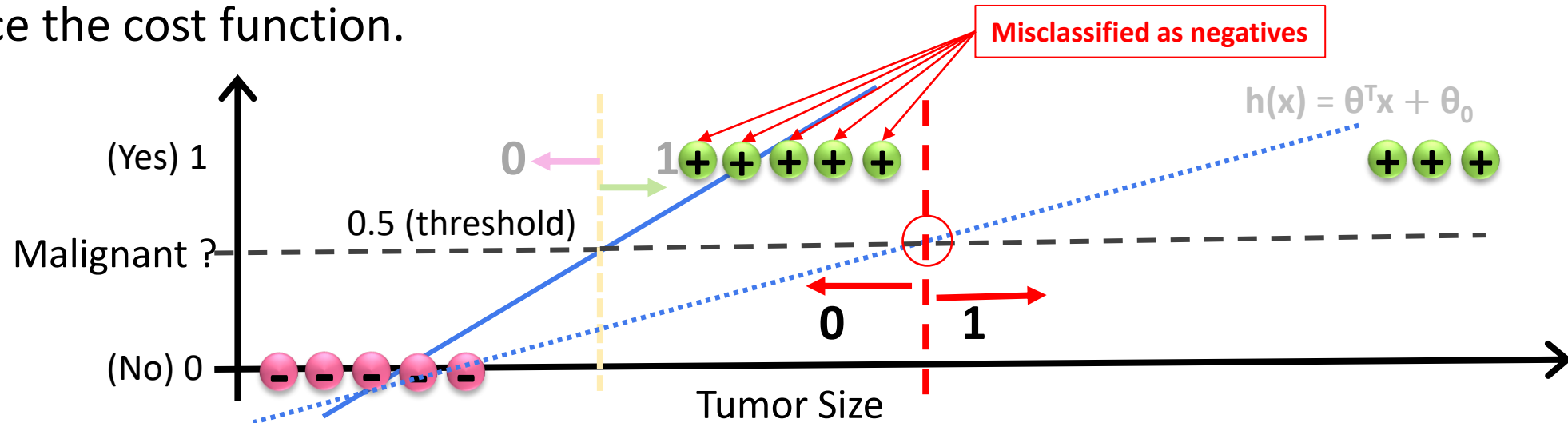
If $h_{\theta}(x) < 0.5$, predict “y = 0”

To predict categories, we can pick a threshold of, say, 0.5.
If the model outputs a value < 0.5 , then $y=0$ or not malignant.
And if the model outputs a value > 0.5 , then $y=1$ or malignant.

Drawbacks of Linear Regression

Consider new 3 data points.

Here the slope of the line changes as the line has to fit the new data point and reduce the cost function.



Threshold classifier output $h_{\theta}(x)$ at 0.5:

If $h_{\theta}(x) \geq 0.5$, predict "y = 1"

If $h_{\theta}(x) < 0.5$, predict "y = 0"

Logistic Regression

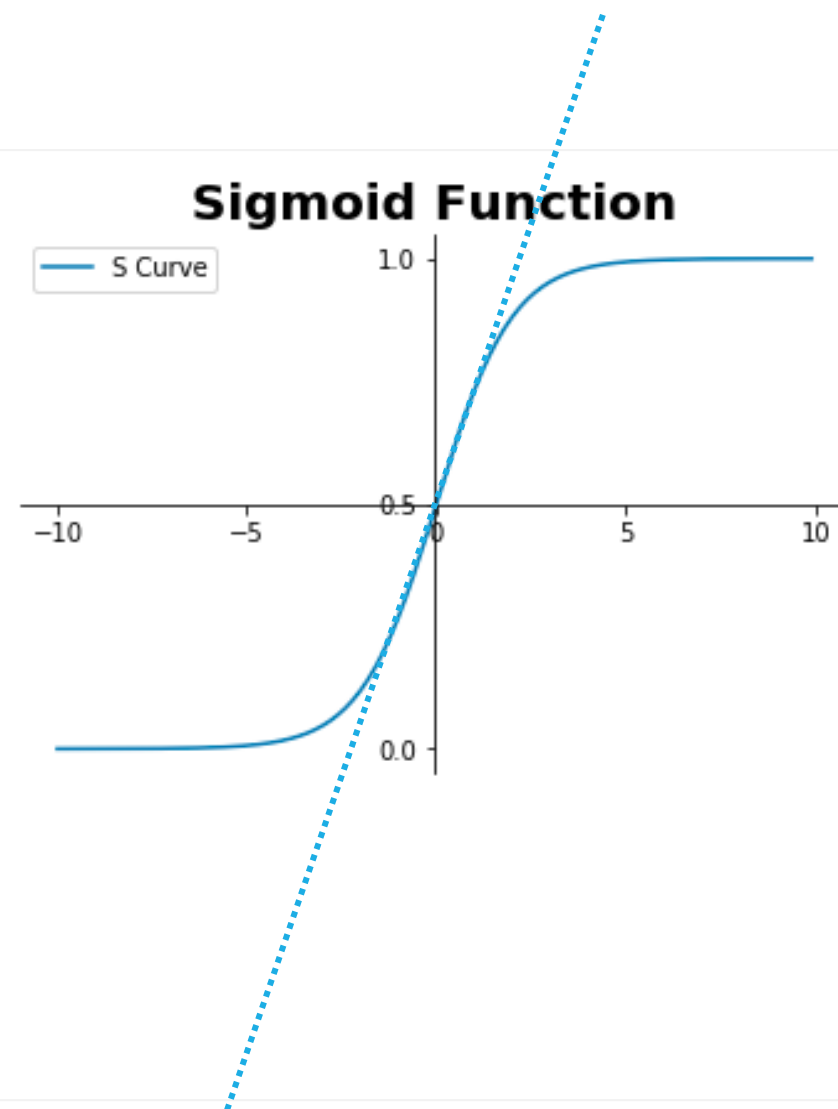
1. Logistic Regression is commonly used to **estimate the probability** that an instance belongs to a particular class.
2. If the estimated probability is **greater than or equal to 50%** then the model predicts that the instance belongs to that class (+) and otherwise it predicts that it does not.
3. Like the Linear Regression model, the Logistic Regression model computes the **weighted sum** of the input features.
4. Instead of giving outputs similar to Linear Regression, the Logistic Regression outputs the **logistic** of these results.
5. The **logistic is the Sigmoid function** that outputs a number between 0 and 1.

Logistic Regression

Classification: $y = 0$ or 1

$h_{\theta}(x)$ can be > 1 or < 0

Logistic Regression: $0 \leq h_{\theta}(x) \leq 1$



Logistic Regression Hypothesis

Logistic Regression Model

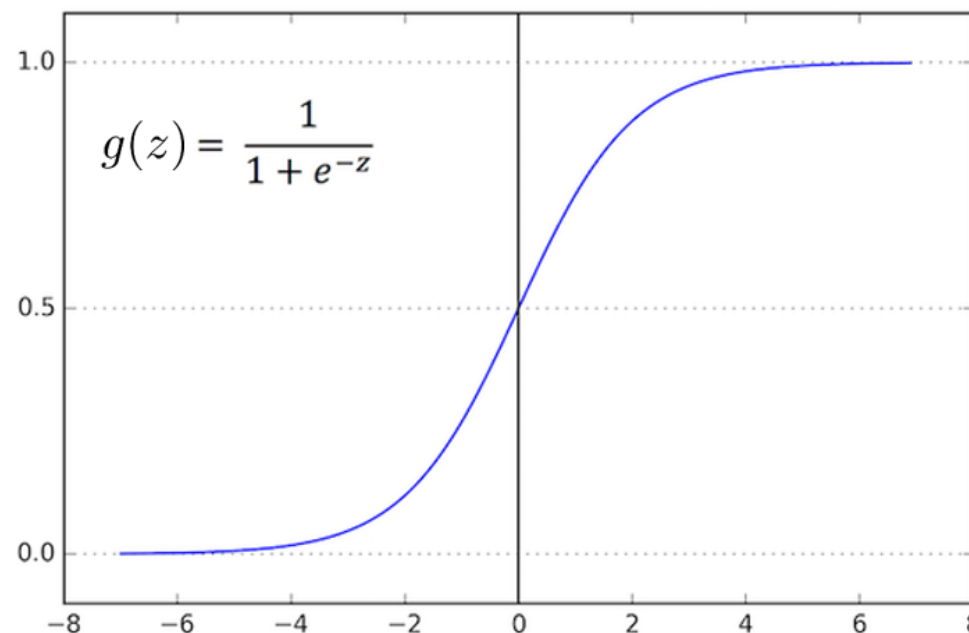
Want $0 \leq h_{\theta}(x) \leq 1$

$$\mathbf{h}(\mathbf{x}) = \theta_0 + \theta_1 \mathbf{x}$$

$$\mathbf{Z} = \mathbf{h}(\mathbf{x})$$

Sigmoid function $\mathbf{g}(\mathbf{Z})$ or Logistic function

- $\mathbf{g}(\mathbf{z})$ = Output between 0 and 1 (probability estimate)
- $\mathbf{z}$ = Input to the function ($\mathbf{Z} = \theta_0 + \theta_1 \mathbf{x}$)
- e = Base of natural log



Logistic Regression Hypothesis

Interpretation of Hypothesis Output

$h_{\theta}(x)$ = estimated probability that $y = 1$ on input x

Example: If $x = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$

$h_{\theta}(x) = 0.7$ Tell patient that 70% chance of tumor being malignant

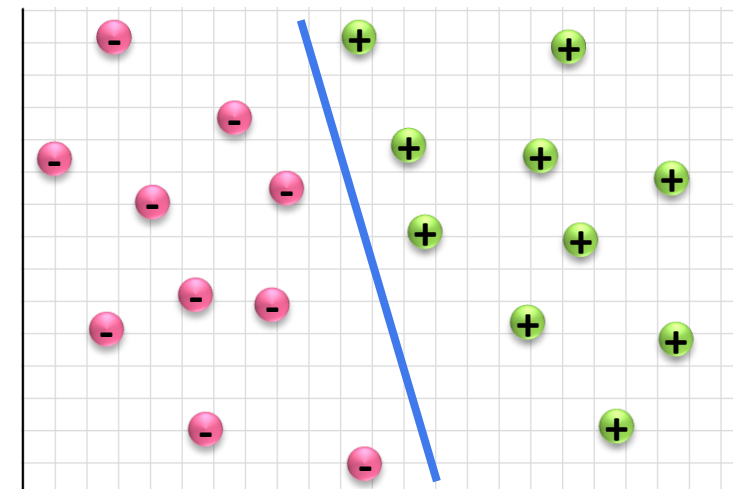
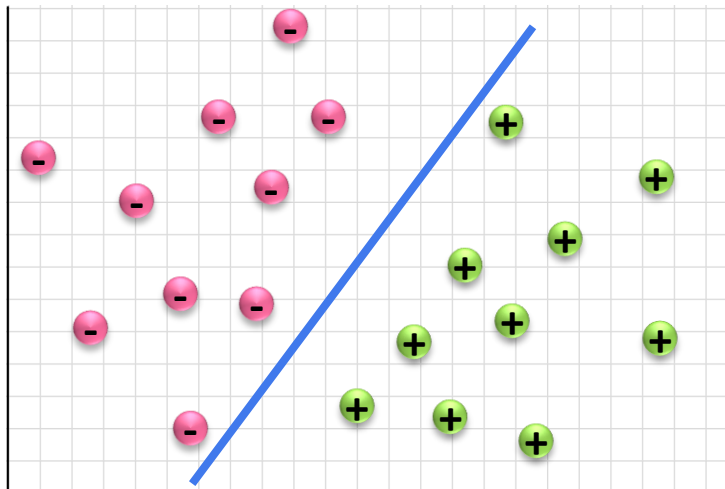
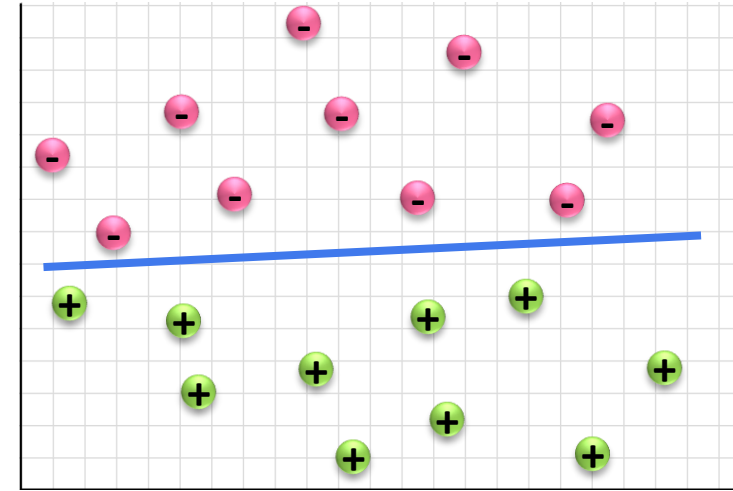
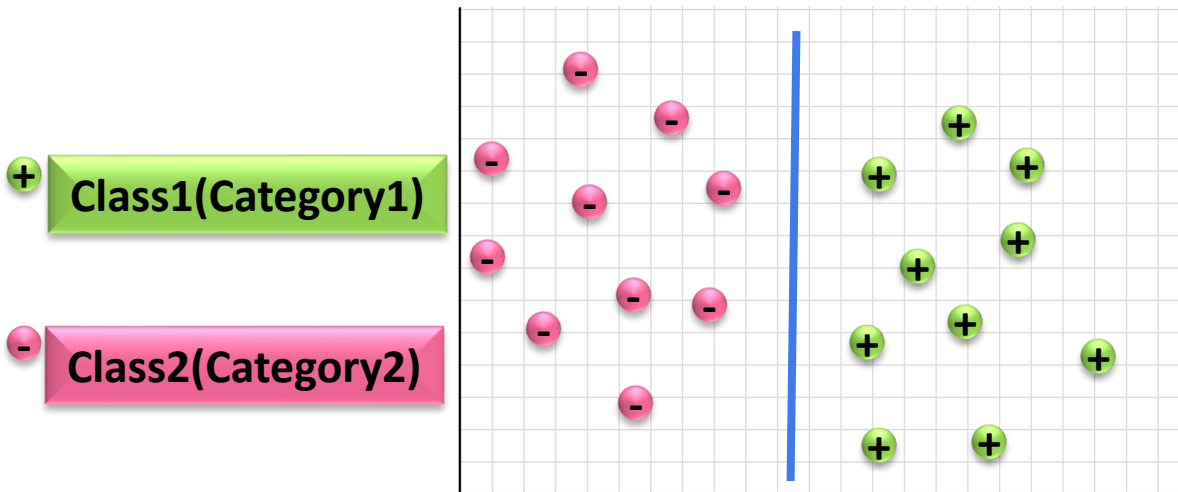
“probability that $y = 1$, given x , parameterized by θ ”

$$P(y = 0|x; \theta) + P(y = 1|x; \theta) = 1$$

$$P(y = 0|x; \theta) = 1 - P(y = 1|x; \theta)$$

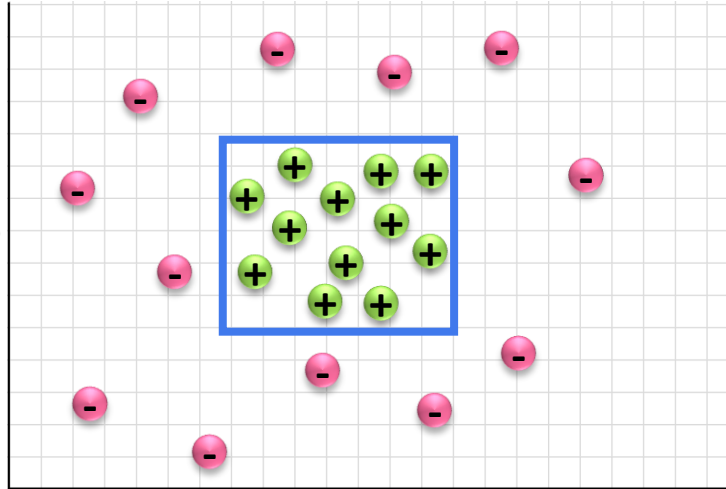
How to find θ ?

Linear Decision Boundaries

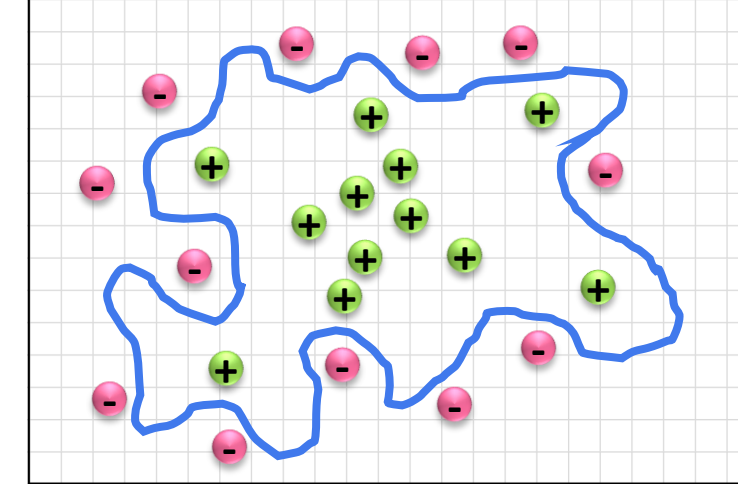
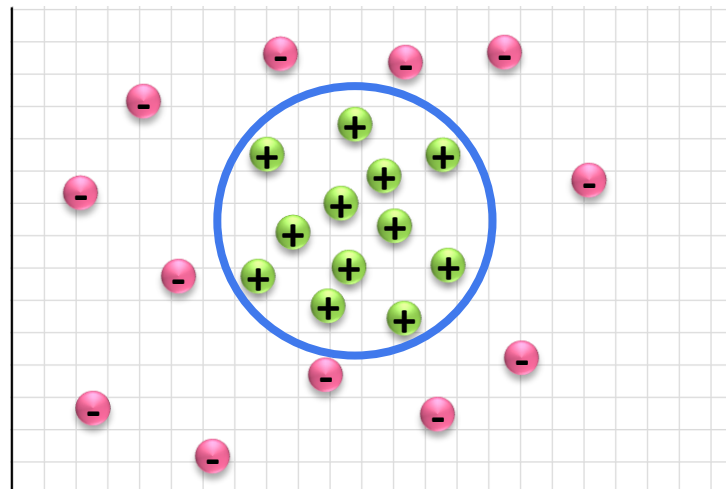
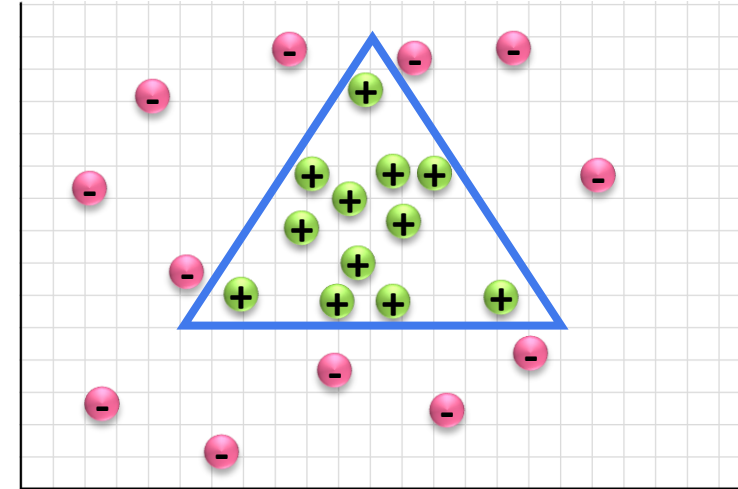


Non-Linear Decision Boundaries

Family Car or
Not a Family Car



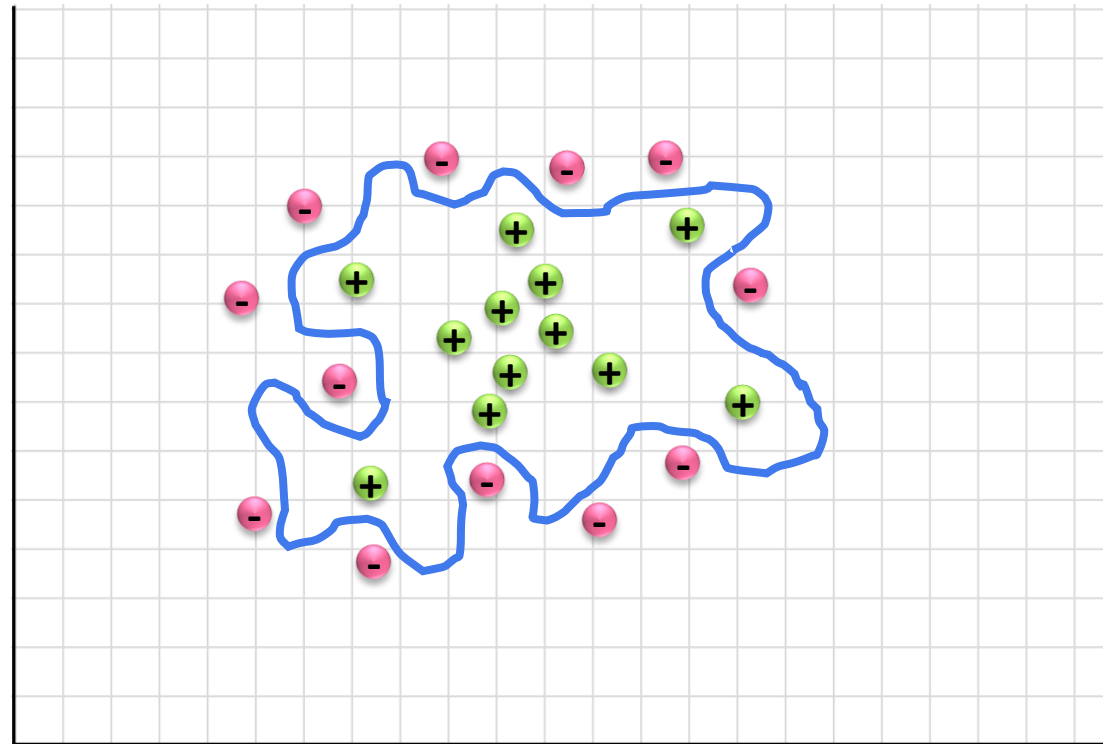
$x_1 = \text{No. of Seats } (\geq 4 \text{ \& } \leq 8)$



Non-Linear Decision Boundary Determination

How the Decision Boundary is Determined?

x_1 = No. of Seats
(≥ 4 & ≤ 8)



Cost Function for Logistic Regression

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

$x^1 \rightarrow \text{Datapoint1(instance)}$

$x_1 \rightarrow \text{Feature1}$

m samples

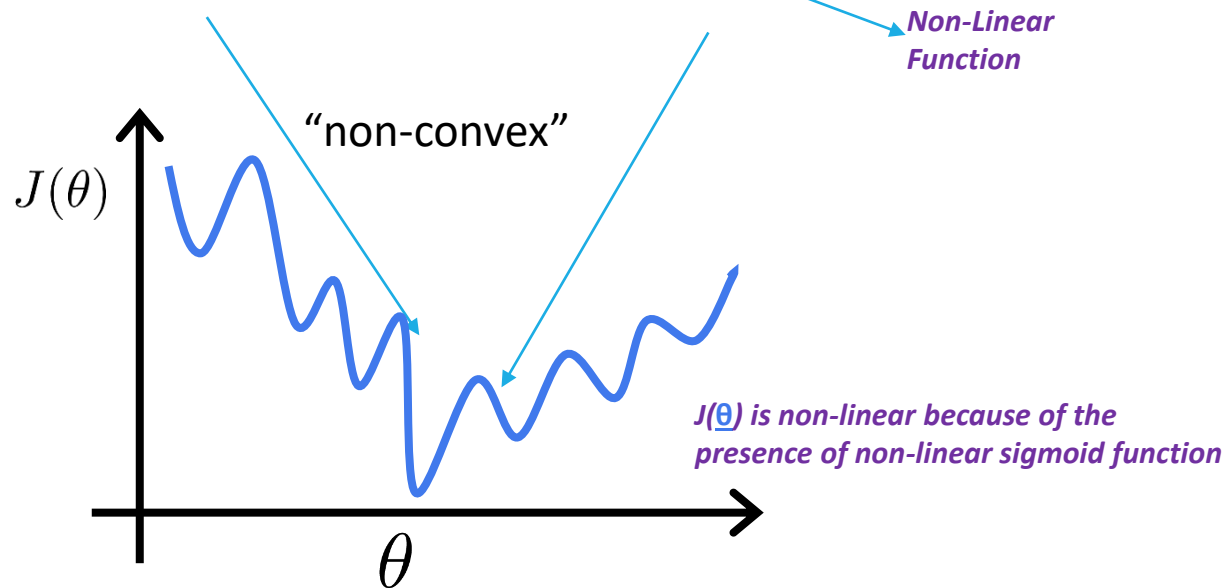
$$x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$$

$$x_0 = 1, y \in \{0, 1\}$$

Cost Function

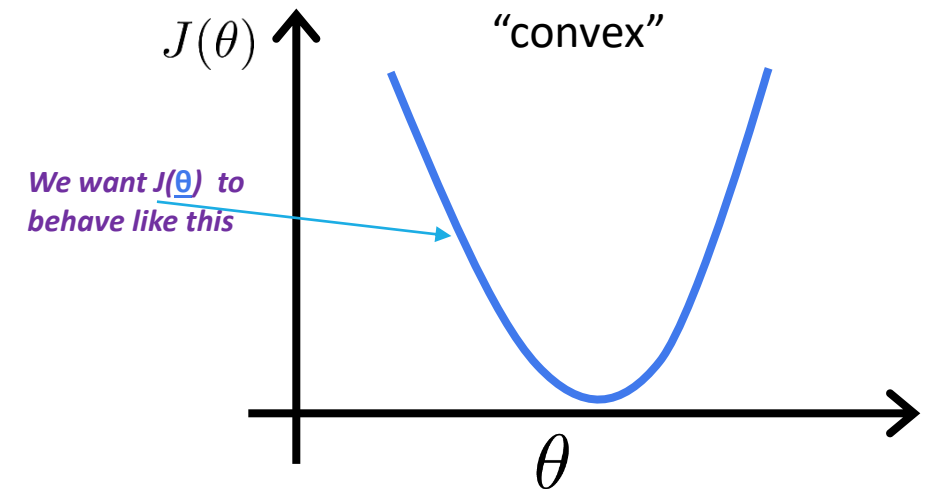
Cost function Logistic Regression:

$$\text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) = \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$



Cost function Linear regression:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (h_{\theta}(x^{(i)}) - y^{(i)})^2$$



Logistic Regression – Training and Cost Function

1. Logistic Regression model **estimates probabilities** and makes predictions.
2. For training the model, setting the parameter vector so that the model estimates **high** probabilities for positive instances $y=1$ and **low** probabilities for negative instances $y=0$
3. This cost function $-\log(z)$ grows **very large when z approaches 0**, so that the cost will be large if the model estimates a probability close to 0 for a **positive instance**.
4. It will also be very large if the model **estimates a probability close to 1** for a **negative instance**.
5. On the other hand $-\log(z)$ is close to 0 when $y=1$, so cost will be close to 0.

Logistic Regression Cost Function

Cost = 0 if $y = 1, h_{\theta}(x) = 1$

But as $h_{\theta}(x) \rightarrow 0$

$Cost \rightarrow \infty$

Captures intuition that if $h_{\theta}(x) = 0$,
(predict $P(y = 1|x; \theta) = 0$), but $y = 1$,
we'll penalize learning algorithm by a very
large cost.

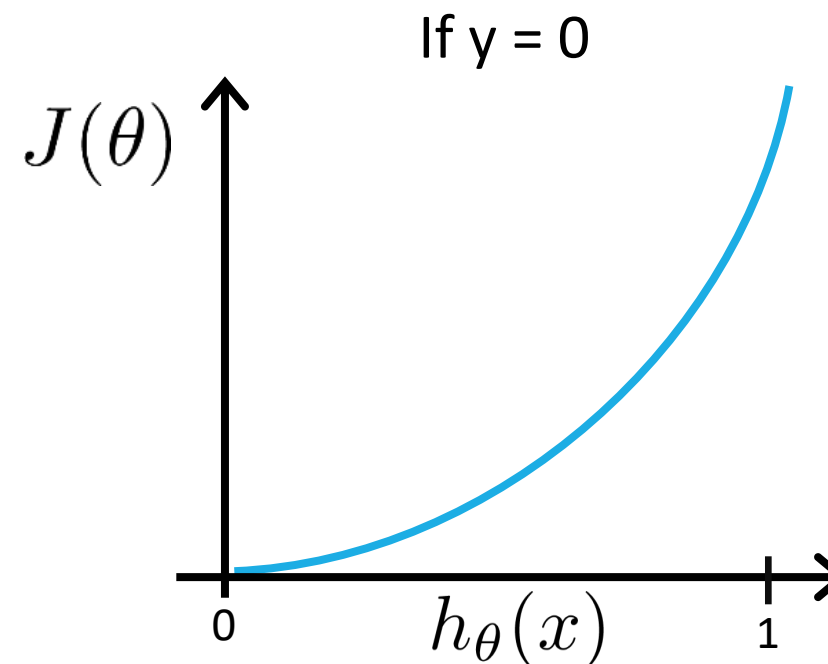
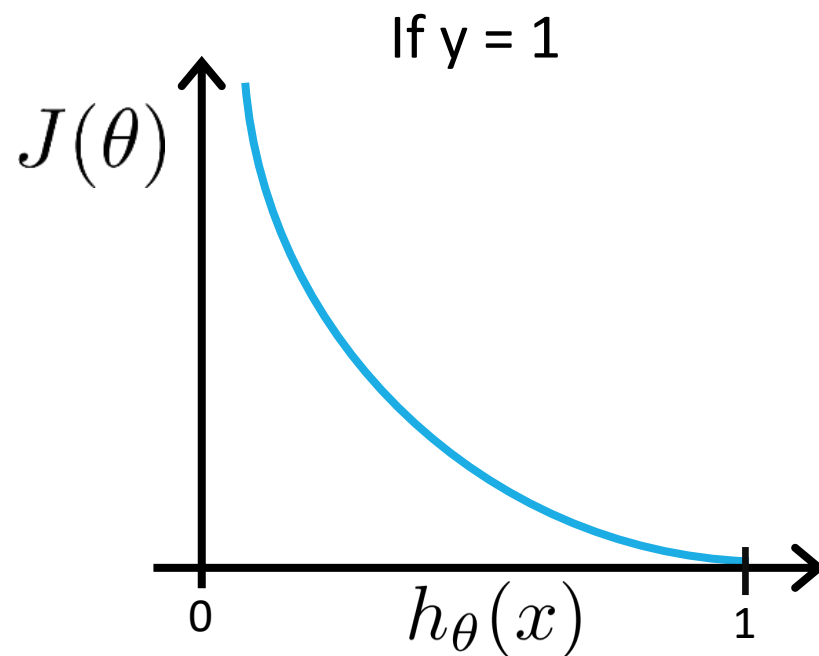
$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Different
Cost Function

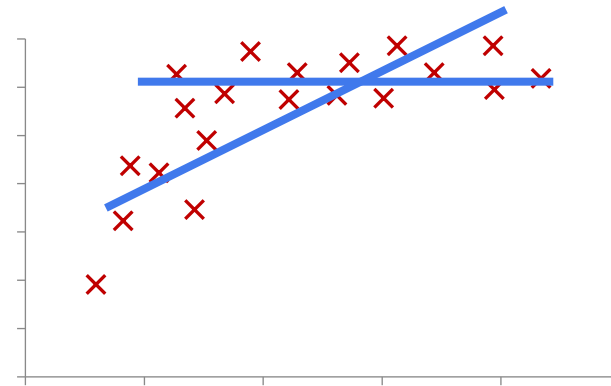
Logistic Regression Cost Function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Observations – Linear Predictors

- Linear Regression assumes a normal distribution of the response variable, which can only be applied to continuous data.
- If we try to build a linear regression model on a discrete/binary y variable, then the linear regression model predicts negative values for the corresponding response variable, which is inappropriate.
- The linear Regression model is not suitable if the relationship between X and y is not linear.



- There exists some non-linear relationship between them. For example, y increases exponentially as X increases.

Non-Linearity for Regression

What happens when we are challenged with **Non-linear data** while using a Linear Regression algorithm?

Can Linear Regression learn patterns from non-linear variables?

How can a straight line fit non-linear data?

To answer all the questions and accommodate **Linear Regression** for a **non-linear dataset** we need a new strategy

This strategy is known as **Polynomial Regression**.

Polynomial Regression

If data is **more complex** then fitting a straight line is not a good approach. In such a case a linear model cannot be used to fit non-linear data.

The approach is to add the powers of each feature as new features, and then train the model on this extended set of features.

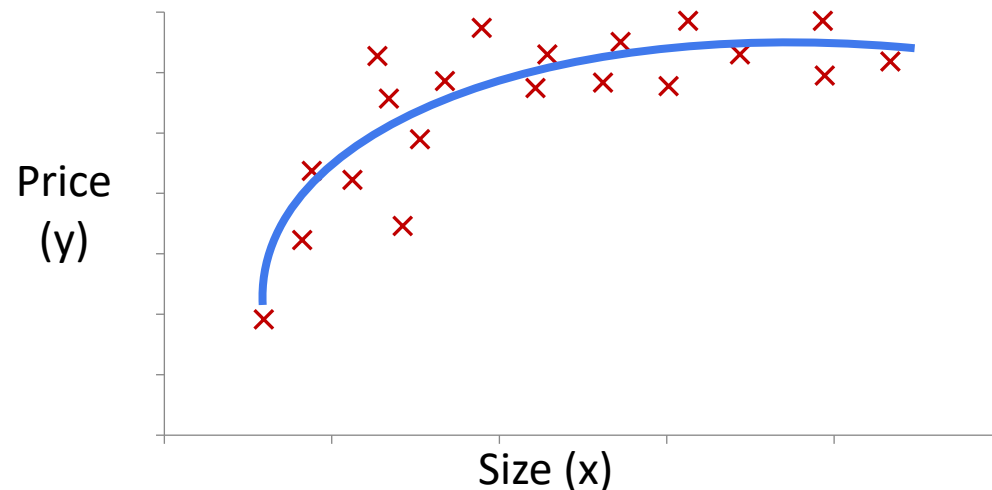
This approach of fitting a linear model to non-linear data is called **Polynomial Regression**.

The process of adding powers to features is known as **Feature Scaling**.

When there are multiple features, Polynomial Regression is capable of finding relationships between the features.

Polynomial Regression

Clearly, a straight line will never fit this non-linear data properly.



$$\theta_0 + \theta_1 x + \theta_2 x^2$$

$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

$$\begin{aligned} h_{\theta}(x) &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \\ &= \theta_0 + \theta_1(\text{size}) + \theta_2(\text{size})^2 + \theta_3(\text{size})^3 \end{aligned}$$

$$x_1 = (\text{size})$$

$$x_2 = (\text{size})^2$$

$$x_3 = (\text{size})^3$$

END
